

Measurement of the Torque on a Single Stretched and Twisted DNA Using Magnetic Tweezers

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We deduced the torque applied on a single stretched and twisted DNA by integrating the change in the molecule's extension with respect to force as it is coiled. While consistent with previous direct measurements of the torque at high forces ($F > 1$ pN), this method, which is simple and does not require a sophisticated setup, allows for lower force estimates. We used this approach to deduce the effective torsional modulus of DNA, which decreases with force, and to estimate the buckling torque of DNA as a function of force in various salt conditions.

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A molecule can be coiled only if it has no bond about which to release the torsion. Thus, most man-made polymers cannot bear torsion since their monomers are linked by a single covalent bond. However, a DNA molecule with no nicks (no breaks in one of the strands) can be twisted, a feature that has important biological implications. First, from a structural point of view, twisted DNA provides an efficient way to compact the molecule so that it fits into the cell or nucleus. Second, a negatively twisted (underwound) DNA may locally denature, thus facilitating the accessibility of its bases to a variety of proteins (RNA polymerases [1], regulation factors [2], etc.). As a result nature has evolved a family of enzymes, generally known as topoisomerases [3,4] that control the torsion and entanglement of DNA molecules. Thus, understanding the behavior of DNA under torsion and estimating the torque arising in a twisted molecule has important biological implications.

Single molecule manipulation experiments [3–5] offer a means to stretch and twist DNA. In these experiments, a DNA molecule is anchored at multiple points (to impede its swiveling) to a surface at one end and to a bead used to apply a force and a torque at the other. In the case of magnetic traps, a superparamagnetic bead is pulled by the field generated by small magnets and twisting is achieved by rotating the magnetic field [6]. With this method the angular position of the bead is imposed and one does not control the applied torque. This setup was nonetheless used to measure the twist-stretch coupling in a DNA molecule via the rotational drag of a small bead attached to the backbone and allowed to swivel to relax the accumulated torsion [7]. More recently, optical tweezers [8] have been used in combination with anisotropic transparent particles to measure the torque from the angular momentum change of a polarized laser beam [9] that also traps the particle. The advantage of the optical tweezers setup is that it allows for a direct measurement of the torque applied on the trapped particle (and through it on the

DNA) as well as the large bandwidth allowing for dynamical measurements. Its drawback is that it involves a rather sophisticated setup which is difficult to use in the low force (below 1 pN) regime that might be more relevant to biological processes.

In this Letter we describe a simple method to estimate the torque applied on DNA by measuring the change in extension of a stretched and coiled molecule as a function of force and number of turns. The magnetic trap system, briefly sketched above and employed extensively in previous experiments [6,10–12], is used to apply a force F on a magnetic bead tethered by a single DNA to a surface and also, by rotating the magnets, to twist the molecule by a known number of turns, n . The results of such experiments are qualitatively easy to understand on the basis of our daily experience with coiling tubes or cords. Consider twisting by n turns a rubber tube of torsional modulus C (usually normalized by $k_B T$ in the DNA context: $C = k_B T C$) held under an applied force F . Initially the torque Γ will increase linearly with n : $\Gamma = 2\pi n C / l$, leaving the extension l almost unchanged. Past a certain number of turns n_b , the associated torque Γ_b becomes so large that it is energetically favorable for the tube to bend rather than to increase its torsional energy: the tube buckles and loops to form a 3D structure called plectoneme, that absorbs torsion as writhe. Further twisting of the tube, while leaving the torque unchanged, results in formation of ever longer plectonemes. Very similar results are observed (see below and Fig. 1) when coiling a DNA molecule: while its extension l varies little for small n it decreases linearly past a certain threshold. As recently suggested [13], from these observations one can deduce the torque applied on the molecule. Indeed the free energy $\mathcal{F}$ of a twisted and stretched molecule depends on the force F and the rotational angle, $\theta = 2\pi n$. The mean extension of the molecule at a given force is: $l = -\partial \mathcal{F} / \partial F|_\theta$ and the mean torque is: $\Gamma = \partial \mathcal{F} / \partial \theta|_F$. One thus readily derives an ex-

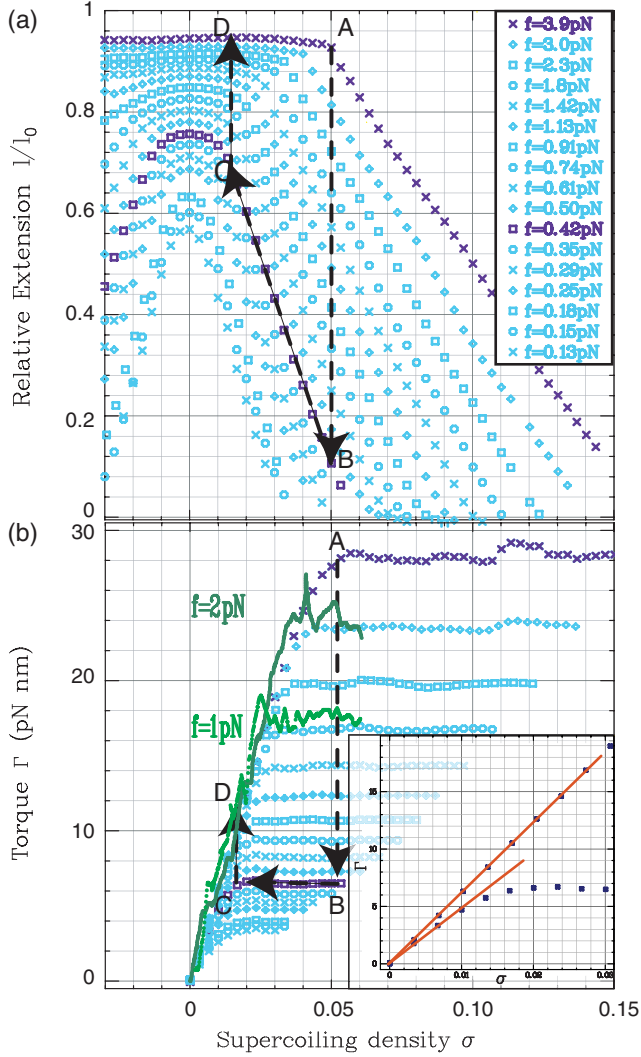


FIG. 1 (color online). (a) Variation of the mean relative extension of a DNA molecule l/l_0 as a function of the degree of supercoiling $\sigma = n/Lk_0$ in 100 mM NaCl. (b) Variation of the torque in DNA as a function of σ . These curves were obtained by integrating Eq. (1) from points A to B and from points C to D in (a) assuming that the torque in the plectonemic regime at a given force (i.e., at points B and C) is constant. The wiggly continuous lines in (b) are results from Ref. [17]. Inset: a zoom-in at low σ displaying two curves (at $F = 0.42$ pN and $F = 3$ pN) with different slope, i.e., different effective torsional modulus.

pression for the mean torque at a given force $\Gamma(F, \theta)$ from a measurement of the decrease in extension with increased coiling $\partial l / \partial \theta|_F$:

$$\Gamma(F, \theta) = \Gamma(F_0, \theta) - \int_{F_0}^F dF' \left(\frac{\partial l}{\partial \theta} \right)_{F'}. \quad (1)$$

Since the angular rotation is known and the force and change in extension with rotation are easily measurable, the determination of the torque difference is reduced to a problem of sampling these variables finely enough to estimate the above integral with sufficient precision. The in-

tegration constant $\Gamma(F_0, \theta)$ is ultimately set by the requirement (see below) that there is no torque on an untwisted molecule: $\Gamma(F, 0) = 0$ (neglecting the small twist-stretch coupling modulus of DNA [7,12]). The integration method presented in this Letter builds upon a similar method introduced earlier [14] which calculated the work done on the DNA molecule by integrating the extension with respect to force at fixed rotation. In the present Letter we integrate the slope of the extension, rather than the extension itself and we are thus insensitive to possible systematic errors in the extension measurement.

With respect to previous similar studies [6,12] we collected a more accurate and extensive set of measurements of DNA's extension (of total length $l_0 \approx 5.4 \mu\text{m}$) in various salt conditions and for different values of force F and degree of supercoiling $\sigma = n/Lk_0$ (the linking number $Lk_0 \approx 1500$ is the number of times the two strands of the molecule wrap around each other), see [15] for the experimental protocol. Figure 1 shows the 100 mM NaCl data series. At low forces ($F < 0.5$ pN) the extension versus rotation curves are symmetric upon $\sigma \rightarrow -\sigma$: the extension is maximal at $\sigma = 0$ and decreases linearly with σ past the buckling threshold (at $\sigma = \sigma_s$). At larger forces ($F > 0.5$ pN) the curve becomes asymmetric as for negative supercoilings DNA denatures before buckling at a critical torque: $\Gamma_d \sim 9$ pN nm [5,14]. For this reason we computed the torque only for positive degrees of supercoiling, although this estimate should also be valid at low forces for negative supercoilings.

To compute the torque at various forces and degrees of supercoiling using Eq. (1), we start from point A at the buckling transition ($\sigma_s = \sigma_A$) which is characterized by the highest stretching force ($F_A = 3.9$ pN) and the highest torque investigated in the data shown in Fig. 1(a). The value of the torque at A (our reference point, initially unknown) is ultimately set by the requirement that $\Gamma(F, 0) = 0$. We can compute the torque at point B by integrating Eq. (1) from F_A to F_B , i.e., along path AB of constant $\sigma = \sigma_A$ but varying force, see Fig. 1(a). The torque in the plectonemic regime is constant: $\Gamma_C = \Gamma_B$. We can thus calculate the torque at point D (at force F_D and $\sigma_D = \sigma_C$) by integrating Eq. (1) along path CD and subtracting the result from the torque difference along path AB. Because this procedure is valid only when points B and C are in the plectonemic regime, it is effectively restricted to values of $\sigma > 0.02$, where the buckling transition can be correctly identified. To evaluate the torque at smaller values of σ , we notice that the values of $\Gamma(F, \sigma)$ at fixed but large forces increase linearly with σ . We therefore extrapolate the values of $\Gamma(F, \sigma)$ for one of these curves down to $\sigma = 0$ requiring $\Gamma(F, 0) = 0$. It does not matter which curve is used: the intercept with the ordinate at $\sigma = 0$ varies by less than 1 pN nm. We then use these extrapolated values to infer from the numerical integration of Eq. (1) the values of the torque at other forces and values

of $\sigma < 0.02$. The results are shown in Fig. 1(b). Various methods to evaluate the derivative $\frac{\partial l}{\partial \theta}|_F$ have been used and found to yield very similar results (a Savitzky-Golay [16] smoothing method was usually preferred as it is less affected by noise due to discrete sampling). The results obtained here are similar, see Fig. 1(b), to the results reported in Ref. [17] (although the shapes of the curves are very similar, they correspond to slightly different force values).

From the data in Figs. 1(a) and 1(b) we can deduce a lower bound for the effective tension and torque on a bare plasmid (circular DNA) unwound by 6% (as often found in nature [18]). The tension is determined by the value of force at which the DNA's extension is zero at $\sigma = 0.06$, i.e., $F \sim 0.42$ pN [19] (the line passing through the points BC in Fig. 1(a) crosses the abscissa at $\sigma = 0.06$). The buckling torque at this force is obtained from Fig. 1(b): $\Gamma \sim 6$ pN nm. This value is close to the value Γ_d where DNA denatures at negative supercoilings [14]. This may explain the variable sensitivity of gene expression to sequence and degree of supercoiling [1].

The slope of the torque vs supercoiling curves at low values of σ [see inset in Fig. 1(b)] yields C_s , i.e., the effective torsional stiffness of DNA, which decreases with the force acting on the molecule (Fig. 2). This was anticipated by Moroz and Nelson [20] who ascribed this variation to a renormalization of the bare torsional stiffness C by thermal fluctuations (that become more impor-

tant at low forces). Their estimate of this effect ($C_s = C[1 - (C/4A)\sqrt{k_B T/AF}]$ obtained by a perturbation expansion at high forces) is shown in Fig. 2. It allows us to deduce a value of $C = 94 \pm 7$ nm, in agreement with previous single molecule estimates of $C \approx 100$ nm [5,17,19]. Other theoretical estimations of C from single molecule force extension data gave values of $C = 75$ nm [21], $C = 86$ nm [14,22], and $C = 109$ nm [23]. The decrease of the effective stiffness C_s at low forces may explain the wide range of values of the torsional constant reported in bulk measurements. Fluorescence polarization anisotropy (FPA) [24] on supercoiled plasmids (effectively at $F \sim 0.4$ pN) yields values as low as $C_s \approx 50$ nm, while both FPA [25] and circularization assays of small DNAs (effectively at large force) yield values of $C_s \sim C = 75$ –100 nm.

While the value of the DNA bare torsional modulus C does not seem to vary much with salt, the buckling torque of DNA Γ_b appears to be much affected by the ionic concentration, increasing by as much as a factor 2 at low salt concentrations. Both Γ_b and the slope of the rotation curve, $\partial l/\partial \theta$, in the plectonemic regime appear to vary as a power law of F , see Fig. 3, an observation for which we have no explanation.

Two recent models describe the behavior of stretched and twisted DNA. The approach of Clauvelin *et al.* [28]

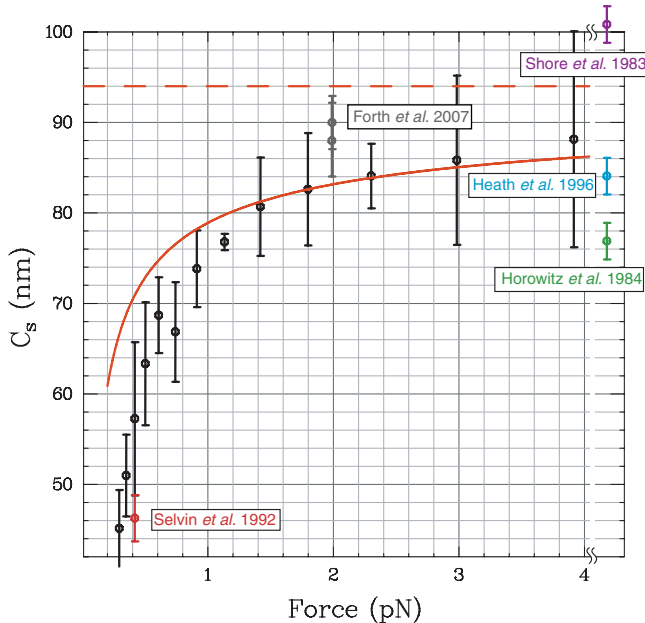


FIG. 2 (color online). Variation with force of the effective torsional stiffness of DNA in 100 mM NaCl. The continuous curve is a best fit of the high force data to the prediction of Ref. [20], with $C = 94 \pm 7$ nm. The figure also displays the values of C_s reported in other single molecule and bulk experiments.

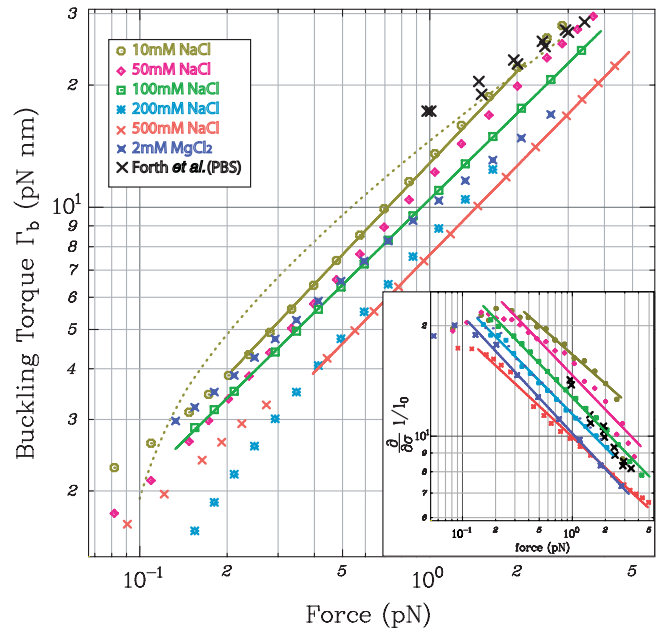


FIG. 3 (color online). Buckling torque of DNA vs force for different salt conditions (10 mM, 50 mM, 100 mM, 200 mM, 500 mM NaCl, 2 mM $MgCl_2$, and data of Ref. [17]). Inset: Variation of the relative slope $\partial(l/l_0)/\partial\sigma$ in the plectonemic regime. Continuous lines: power law fits of the monovalent salt series with exponents: 0.72 ± 0.07 and -0.30 ± 0.03 (inset). Dotted lines: best fit to the model of Ref. [30] for the 10 mM NaCl series.

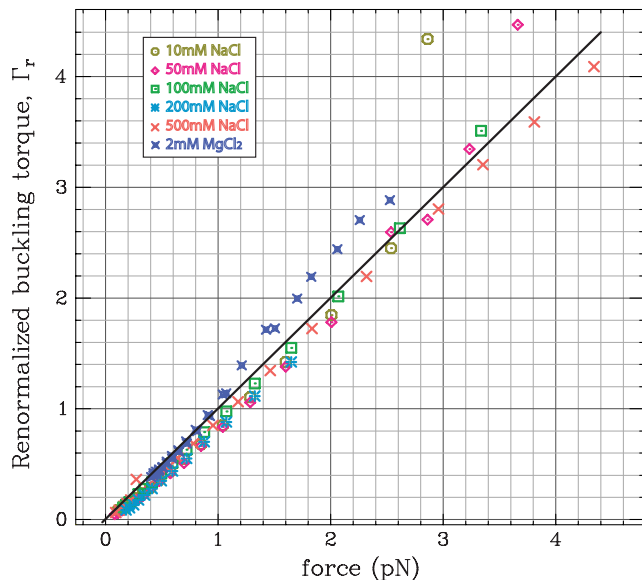


FIG. 4 (color online). Variation of the normalized torque $\Gamma_r = (3\pi\Gamma_b Lk_0 z)/(2\partial l/\partial\sigma)$ vs force for the same buffer conditions of Fig. 3. The NaCl series collapses on the continuous line predicted from Ref. [28], while the MgCl_2 series yields values a little higher than expected. The two high force points that depart from the diagonal are due to the abrupt change of slope at the transition to *P*-DNA [31].

treats DNA as a classical mechanical tube of relative extension $z = l(\sigma = 0)/l_0$ given by the wormlike chain model [19,29]. It predicts that the renormalized buckling torque $\Gamma_r = \alpha\Gamma_b$ [with $\alpha = (3\pi Lk_0 z)/(2\partial l/\partial\sigma)$] is equal to the force F . As seen in Fig. 4, this relation is remarkably satisfied by the data obtained at various forces and in different buffers.

Marko [30] proposed a more consistent statistical mechanics treatment of stretched and twisted DNA. In this model the molecule is partitioned between an unstretched plectonemic supercoiled phase with torsional stiffness P and a stretched and twisted DNA molecule with persistence length $A = 50$ nm and effective torsional stiffness C_s . It is, however, impossible to fit our data over their whole force range to this model with a single fit parameter, P (see Fig. 3). In some sense this is not surprising since the description of the plectonemic phase is oversimplified in this model: for example, it does not take into account the variation of plectonemic radius with force due to entropic repulsion [29].

In conclusion, we presented a novel method for obtaining the torque applied on a single DNA molecule from extension measurements at equilibrium. The method can be used to deduce the mechanical properties (torsional stiffness, buckling torque, etc.) of other DNA fibers (e.g., RecA coated DNA, chromatin, etc.). Because it is based on

equilibrium thermodynamic relations of DNA under tension and torsion it is in general not suitable for dynamical studies of DNA interactions with proteins, except in the plectonemic regime where the buckling torque (that we can measure in absence of protein) is constant.

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